Fiscal Year:	FY 2005	**Task Last Updated:**	FY 07/19/2005
PI Name:	Spaepen, Frans Ph.D.		
Project Title:	Glass Formation, Nucleation and Diffusion in Metallic Melts		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Metals and alloys		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	02138-2901	**Congressional District:**	8
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	98-HEDS-05
Start Date:	06/01/2000	**End Date:**	11/30/2004
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	0	**No. of Master' Degrees:**	1
No. of Master's Candidates:	3	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA MSFC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):	Turnbull, D. (Harvard University) Aziz, M. J. (Harvard University)		
Grant/Contract No.:	NAG8-1680		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Diffusivity in liquid metals, solutes and diffusion in Ni and Si. Nucleation of silicon crystals from the melt following melting by pulsed laser irradiation. The aim is to improve the experiments of Stifler et al. by measuring the nucleation temperature directly, rather than to estimate it from heat flow. Better knowledge of this temperature is essential for comparison to nucleation experiments in bulk liquid Si, either by levitation (Herlach's group at DLR) or fluxing (our earlier work). Such a comparison will give a new value for the temperature coefficient of the crystal-melt interfacial tension; this value, in turn, has implications for the structure of the interface. Thermodynamics and structure of the crystal-melt interface. The crystal-melt interface in silicon is of direct interest to the experimental study of part 2. We are also studying the interface between polytetrahedral crystals and their melt. Fabrication of metallic glass foam by consolidation of glass powders and foaming agents. The objective is to develop a basic understanding of the phenomena		

	and processes that are central to the microgravity program: crystal nucleation and growth, glass formation, the structure and thermodynamics of the crystal-melt interface, and diffusion in the liquid state. Studies of the undercooling of liquids and the kinetics of crystal nucleation are an important category of experiments that exploit the containerless environment provided by microgravity. Our ground-based work is aimed at advancing the understanding of the fundamentals of undercooling, nucleation, and glass formation, at exploring the potential and limitations of ground-based alternatives such as fluxing and drop-tube processing, and at exploring the potential of ground-based containerless processing facilities provided by the microgravity program.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	We discovered that undercooled metallic melts of GaAs based alloys lead to rapid solidification of metastable phases with unique electronic and magnetic properties, in collaboration with O.D. Dubon and colleagues at LBNL. We investigated the formation of the highly mismatched alloy GaN_xAs_1-x using a combination of pulsed laser melting and rapid thermal annealing. This material forms quantum wells when surrounded by GaAs. Thin films with unusually high nitrogen content and nigh nitrogen activation efficiency (up to 50%) were synthesized; their structural and optical properties are comparable to films grown by epitaxial deposition techniques with similar substitutional nitrogen content. The effects of nitrogen implantation dose, laser energy fluence and rapid thermal annealing temperature on the nitrogen incorporation as well as optical and structural properties of the GaNxAs1-x films have been mapped out. We also demonstrated the formation of ferromagnetic Ga1-xMnxAs films by Mn ion implantation into GaAs followed by pulsed laser melting. These materials have remanent magnetization and Curie temperature comparable to those in thin films grown by molecular beam epitaxy. Our method, however, using pulsed laser melting and rapid solidification, has a greater flexibility for lateral patterning. These results have been reported in three archival journal publications.
Bibliography Type:	Description: (Last Updated: 06/29/2005)
Articles in Peer-reviewed Journals	Yu, K.M.; Walukiewicz, W.; Beeman, J.W.; Scarpulla, M.A.; Dubon, O.D.; Pillai, M.R.; Aziz, M.J. "Enhanced nitrogen incorporation by pulsed laser melting of GaNxAs1-x formed by N implantation" Applied Physics Letters, 80:3958-3960 , Jan-2003
Articles in Peer-reviewed Journals	Scarpulla, M.A.; Dubon, O.D.; Yu, K.M.; Monteiro, O.; Pillai, M.R.; Aziz, M.J.; Ridgway, M.C. "Ferromagnetic Ga1-xMnxAs films produced by ion implantation and pulsed laser melting" Applied Physics Letters 82:1251-1253 , Jan-2003
Articles in Peer-reviewed Journals	Yu, K.M.; Walukiewicz, W.; Scarpulla, M.A.; Dubon, O.D.; Wu, J.; Jasinski, J.; Liliental-Weber, Z.; Beeman, J.W.; Pillai, M.R.; Aziz, M.J. "Synthesis of GaNxAs1-x thin films by pulsed laser melting and rapid thermal annealing of N+-implanted GaAs" Journal of Applied Physics 94:1043-1049 , Jan-2003
Articles in Peer-reviewed Journals	Holland-Moritz, D.; Spaepen, F. "The magnetic contribution to the driving force for crystal nucleation in undercooled Co-Pd melts" Philosophical Magazine 84:957-966 , Jan-2004
Articles in Peer-reviewed Journals	Asta, M.; Spaepen, F.; van der Veen, F. "Solid-liquid interfaces: Molecular structure, thermodynamics and crystallization" MRS Bulletin, pp. 920-926 , Dec-2004
Articles in Peer-reviewed Journals	Wu, D.T.; Granasy, L.; Spaepen, F. "Nucleation and the solid-liquid interfacial free energy" MRS Bulletin, pp. 945-950 , Dec-2004
Articles in Peer-reviewed Journals	Leonard, J.P.; Aziz, M.J.; Renk, T.J.; Thompson, M.O. "Solute diffusion in liquid nickel measured by pulsed ion beam melting" Metallurgical and Materials Transactions A 35:2803-2807 , Dec-2004
Awards	Spaepen, F. "External Member, Max Planck Society" Jan-2003
Awards	Spaepen, F. "Max Planck Lecturer, Max Planck Institutes, Stuttgart" Dec-2004
Awards	Spaepen, F. "J.B. Cohen Lecturer, Northwestern University" Oct-2004
Awards	Aziz, M.J. "Fellow, AAAS" Jan-2004
Books/Book Chapters	Aziz, M.J. "Tests of theories for nonplanar growth during rapid alloy solidification" in "Solidification Processes and Microstructures: A Symposium in Honor of Wilfried Kurz" Ed. Rappaz, M., et al. TMS, Jan-2004
Dissertations and Theses	Warrender, J.M. "Morphological Evolution of Nanocrystal Metal-on-Insulator Films Grown by Pulsed Laser Deposition" Division of Engineering and Applied Sciences, Harvard University Jun-2004
Papers from Meeting Proceedings	Holland-Moritz, D.; Spaepen, F. "Magnetic effects on nucleation in undercooled Co-Pd melts" Proceedings of the European Congress on Advanced Materials and Processes Dec-2003
Presentation	Spaepen, F. "Energies and entropies in materials science" J.B. Cohen Lectures, Department of Materials Science and Engineering, Northwestern University Oct-2004
Presentation	Spaepen, F. "The structure of liquids and solid-liquid interfaces" J.B. Cohen Lectures, Department of Materials Science and Engineering, Northwestern University Oct-2004

Presentation	Spaepen, F. "Mechanical properties of metallic glasses" J.B. Cohen Lectures, Department of Materials Science and Engineering, Northwestern University Oct-2004
Presentation	Aziz, M.J. "Nonplanar growth during rapid alloy solidification" University of Catania Feb-2004
Presentation	Aziz, M.J. "Tests of theories for nonplanar growth during rapid alloy solidification" TMS-AIME Symposium on Solidification Microstructures: A Symposium in Honor of Professor Wilfried Kurz, Charlotte, N.C. (invited) Mar-2004